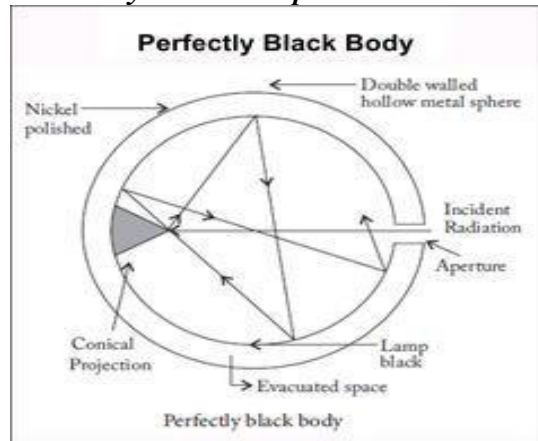


Blackbody Radiation:

A Blackbody is one which absorbs all the light which is incident on it and re-radiates all those absorbed radiations when it is heated to incandescence (red hot).

Blackbody Radiation spectrum:

A Blackbody is one which absorbs the entire radiation incident on it and emits all the absorbed radiation when it is hotter. A true blackbody does not exist practically. Many Scientists tried to devise an ideal blackbody, however, a blackbody designed by Wien has features very close to the true blackbody. A blackbody at a particular temperature found to emit a radiation of all possible wavelengths. The radiations emitted are analysed and is called blackbody radiation spectrum.

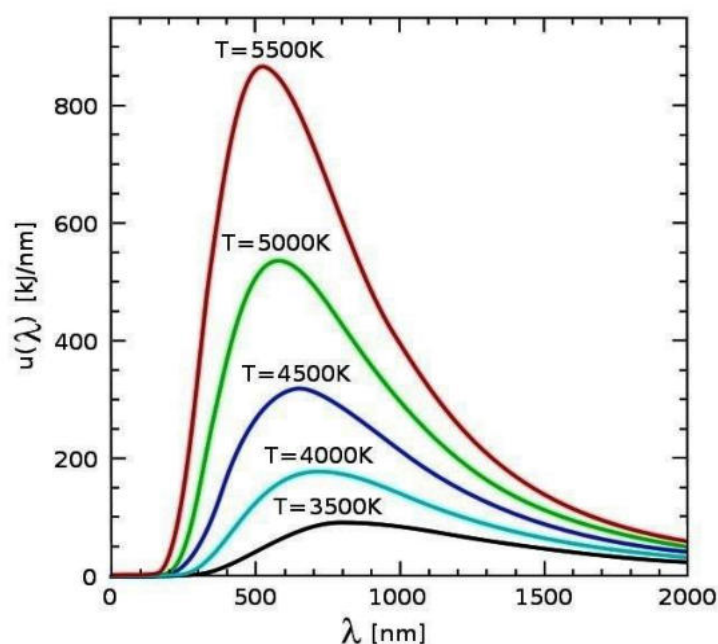
Nature of Blackbody radiation Spectrum: The radiations emitted is analysed and the intensity (or the energy density) of emitted radiation is plotted against the wavelength as shown in fig.

(i) At a constant temperature, radiations of different wavelengths are emitted out in different quantities. It is a continuous spectrum starting from certain minimum wavelength to maximum wavelength. There exist a particular wavelength (Called λ_m) for which intensity of emitted radiation is maximum.

(ii) There are different curves for different temperatures. The intensity emitted radiation increases with the increase of the temperature.

(iii) As the temperature of the body increases, the wavelength corresponding to maximum intensity (λ_m) shifts towards lower wavelength side.

(iv) The area under the curve gives the total radiant energy. The area under the curve is proportional to the forth power of the absolute temperature. This is called Stefan's 4th power law [$E \propto T^4$].

**Planck's Law:**

In 1900, Max Plank explained the entire blackbody radiation spectrum by introducing the revolutionary idea of quantization of energy of the constituent atoms of the blackbody.

Assumptions:

i) A blackbody consists of large number of electrical oscillators whose energy is quantized. An oscillator may have any integral value for its energy given by $E = nh\nu$, where n is any positive integer, h is plank's constant, ν is the frequency of oscillation.

- ii) An oscillator may gain or lose energy by absorbing or emitting a radiation of frequency ν respectively where $\nu = \frac{\Delta E}{h}$, ΔE is difference in energies of the oscillator before and after the emission or absorption take place.

With these assumptions Planck at an expression for distribution of energy of blackbody radiation as The energy radiated per unit volume from the surface of the blackbody in the wavelength interval λ and $\lambda + d\lambda$ is

$$E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\left(\frac{hc}{\lambda kT}\right)} - 1} d\lambda \quad \text{----- (1)}$$

This is Planck's Law of Blackbody Radiation. The values predicted by Planck's law fitted with the experimental values and hence explains the nature of blackbody radiation completely.

Note: Planck's law signifies the particle nature of blackbody radiation. With the help of this, Einstein explained Photoelectric effect. Later Compton used the same idea to explain Compton Effect of the X rays. These two phenomena lead to assume dual nature for Electro Magnetic Radiation.

Matter waves

De Broglie Hypothesis (Wave particle dualism)

Some of the Phenomena observed around 1900, like blackbody radiation, photoelectric effect etc, can be explained by assuming the particle nature. The phenomena like interference diffraction, polarization etc, can be explained only by assuming wave nature. Therefore it can be concluded that neither the wave theory nor the particle theory can be disputed and both are equally valid and light exhibit dual nature.

Nature manifest herself mainly in two forms, one is the energy and the other is matter. If light being a form of energy exhibit dual nature, then why cannot the matter? Based on these symmetry arguments, one can expect the matter also to exhibit dual nature.

Based on these symmetry arguments, in 1923, Luis de Broglie argued that "every moving particle is associated with wave nature, whose wavelength is given by $\lambda = \frac{h}{p} = \frac{h}{mv}$, Where h is the Planck's constant, m is the mass, v is the velocity and p is the momentum of the moving particle". This statement (a logical argument without experimental proof) is called de Broglie hypothesis.

Such waves associated with moving particle are called as matter waves or de Broglie waves, and the wavelength $\lambda = \frac{h}{p} = \frac{h}{mv}$ is called de Broglie wavelength.

Properties of Matter waves

- 1) Matter waves are the waves associated only with the moving particle. The particles at rest cannot exhibit the wave nature.
- 2) Matter waves are called as pilot waves or guiding waves; they move with a velocity greater than the velocity of light leaving the particle behind.
- 3) Matter waves are also called as the probability waves; the amplitude of matter waves reveals the probability of finding the particle at a point in a given region of space.
- 4) Unlike EM waves, the matter waves of different wavelengths travel with different phase velocities.

de Broglie wavelength in different forms:

- The de Broglie wavelength of a particle of mass 'm' moving with a velocity 'v' and momentum 'p' is given by $\lambda = \frac{h}{p} = \frac{h}{mv}$

Using the classical equation $p = mv$,

We get $p^2 = m^2 v^2 = 2m \left(\frac{1}{2} m v^2 \right) = 2m E_k$

Or $p = \sqrt{2mE_k}$

∴ The de Broglie wavelength of a particle moving with a kinetic energy E_k is

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

- For an electron accelerated by a potential difference of V volts, the kinetic energy is $E_k = qV$.

The de Broglie wavelength of an electron accelerated by a potential difference of V volts is given by

$$\lambda = \frac{h}{\sqrt{2meV}}.$$

$$\text{Note: } \lambda = \frac{h}{\sqrt{2meV}} = \frac{1}{\sqrt{V}} \left[\frac{6.625 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19}}} \right] = \frac{1.226 \text{ nm}}{\sqrt{V}}$$

- The de Broglie wavelength of a particle which is in thermal equilibrium at a temperature T is

$$\lambda = \frac{h}{\sqrt{3mKT}}$$

Where K is the Boltzmann's constant $K = 1.38 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$.

Note:

The velocity of the de Broglie waves calculated by classical approach exceeds the velocity of light. At that time, it was believed that the velocity of light is the maximum attainable velocity. This contradictory was resolved by Max Born by suggesting a moving particle can associate with number of de Broglie waves whose wavelength is likely to be in the neighbourhood of that predicted by de Broglie. According to Max Born, all such coherent waves will interfere constructively over a limited region of space where the particle is located (to form a wave packet) and interfere destructively elsewhere. The individual de Broglie waves move with a velocity greater than the velocity of light leaving the particle behind. This velocity is called 'Phase Velocity'. The wave packet enclosing the particle moves with a different velocity called 'Group Velocity'.

According to Max Born, the particle can be located anywhere inside the wave packet. Hence there is an uncertainty regarding its position. Meanwhile if there exist no de Broglie waves associated with the same particle, there is an uncertainty in its momentum (because $= \frac{h}{p}$). This is known as uncertainty principle.

Heisenberg's Uncertainty Principle:

Statement: "In any simultaneous measurement of position and momentum of a moving particle, the product of the corresponding uncertainties, which is inherently present in the measurement, is always greater than or equal to $\left(\frac{h}{4\pi} \right)$ ".

$$\text{ie. } \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Where Δx is the uncertainty regarding the position and Δp is the uncertainty regarding the momentum.

Explanation: If position is measured with great accuracy, then the simultaneous measurement of the momentum becomes less accurate, and vice versa. The product of the two corresponding uncertainties is always constant and greater than the limit $\left(\frac{h}{4\pi} \right)$.

Note: Such pair of variable which cannot be measured simultaneously with great accuracy are called canonically conjugate variables. The above uncertainty relation is given for 1-D motion. For the 3-D motion the uncertainty relation is

$$\begin{aligned}\Delta x \cdot \Delta P_x &\geq \frac{h}{4\pi} \\ \Delta y \cdot \Delta P_y &\geq \frac{h}{4\pi} \\ \Delta z \cdot \Delta P_z &\geq \frac{h}{4\pi} .\end{aligned}$$

Such pair of variable which cannot be measured simultaneously with great accuracy are called canonically conjugate variables. The other examples of canonically conjugate variables are the **angular momentum and angle** as well as **energy and time**.

ie. $\Delta L \cdot \Delta \theta \geq \frac{h}{4\pi}$

Where ΔL is the uncertainty in the measurement of angular momentum

And $\Delta \theta$ is the the uncertainty in the measurement of angular position

and $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$

Where ΔE is the uncertainty in the measurement of energy

And Δt is the the uncertainty in the measurement of time

Application of H U P:

(To prove the nonexistence of electron inside the nucleus using HUP)

Consider that the electron is present inside the nucleus.

If an electron is present inside the nucleus, then the error in the determination of its position

$$\Delta x \leq 1 \times 10^{-14} \text{ m}.$$

Then using HUP, $\Delta p \geq \frac{h}{4\pi \cdot \Delta x} = \frac{6.625 \times 10^{-34}}{4\pi (1 \times 10^{-14})} = 5.27 \times 10^{-21} \text{ kg} \cdot \text{ms}^{-1}.$

$\therefore \Delta p \cong p = 5.27 \times 10^{-21} \text{ NS}$

The energy of the electron is given by classical equation $E = \frac{p^2}{2m}$

$$E \geq \frac{(5.27 \times 10^{-21})^2}{(2 \times 9.1 \times 10^{-31})} = 1.5 \times 10^{-11} \text{ Joul.}$$

Or $E \geq 95.37 \times 10^6 \text{ eV} \cong 95 \text{ MeV}.$

I.e. If the electron resides inside the nucleus, it should have a minimum energy of 95Mev. But it has been experimentally verified that no electron of an atom do not possess that much of huge energy. Even the β particles (which are most energetic electrons) possess a maximum energy of nearly 4Mev. This contradictory result proves the nonexistence of electron inside the nucleus.

Wave function

Waves, in general, are associated with some quantity that varies periodically. For example, water waves involve the periodic variation of the height of water surface at a point. Similarly, sound waves are characterised by the periodic variation of pressure. The periodic variation of mutually perpendicular electric and magnetic vector characterises the electromagnetic waves. In the same way, 'the quantity that characterises the matter waves is called wave function'. It is always denoted by the Greek letter Ψ . The wave function contains all the information about the dynamical behaviour of the particle exhibiting the wave nature.

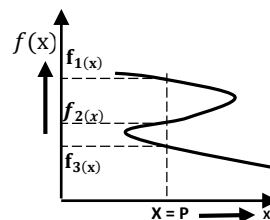
A wave function is also a solution Schrodinger's wave equation which describes the dynamical behaviour of a moving particle. As Schrodinger's wave equation is a second order differential equation, the wave function is complex in nature.

Properties of Wave function:

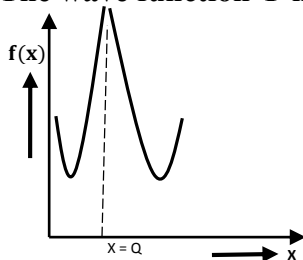
There may be a number of solutions for the Schrodinger's wave equation. Only the wave functions which satisfy the properties given below are called acceptable wave functions.

1) The wave function Ψ should be single valued everywhere.

Consider an example $f(x)$ as shown in figure. At $X = P$, the function $f(x)$ has three different values. If $f(x)$ were to be the wave function, the probability of finding the particle at $X = P$ has three different values which is absurd. Hence the function $f(x)$ cannot be the wave function.



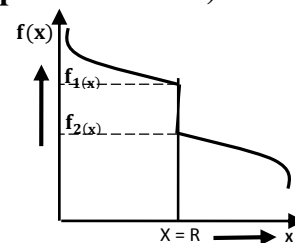
2) The wave function Ψ must be finite everywhere.



Consider a function $f(x)$ as shown in figure. At $X = Q$, the $f(x) = \infty$. $f(x)$ were to be the wave function, the probability of finding the particle at $X = Q$ cannot be ∞ . Therefore $f(x)$ cannot be the wave function

3) The wave function Ψ and its first derivatives (both in space and time) must be continuous everywhere.

Consider a function $f(x)$ as shown in figure. The function $f(x)$ is not continuous at $X = R$, L.H.L. $\neq$ R.H.L. The function $f(x)$ may assume any value between $f_1(x)$ and $f_2(x)$ as it is not defined properly. In other words $f(x)$ has a discontinuity at $X = R$. As $f(x)$ is not continuous, its first derivatives cannot be finite. Therefore $f(x)$ cannot be a wave function.



4) For the bound state, the wave function must vanish at infinity. If the wave function is complex, then the quantity $\Psi^* \Psi dx$ must vanish at infinity.

Physical Significance of wave function

- Born's Interpretation (Probability density):

In Quantum Mechanics, it is postulated that the wave function contain all the information about the moving particle exhibiting wave nature. As the wave function is complex in nature, the quantity ' Ψ ' has no direct physical significance. However, when it is multiplied with its complex conjugate Ψ^* , it would yield a real meaning. The quantity $\Psi^* \Psi = |\Psi|^2$ is called the probability density. The probability density is a measure of probability of finding the particle at point in a given region of space.

The probability of finding the particle over an interval dx is given by $\Psi^* \Psi dx = |\Psi|^2 dx$

The probability of finding the particle over entire space (1D) is given by

$$\int_{-\infty}^{\infty} \Psi^* \Psi dx = \int_{-\infty}^{\infty} |\Psi|^2 dx$$

Normalization:

For a single particle system, the probability density should be always unity. Consider an example that we are trying to calculate the probability of finding an electron at a point on the Bohr's orbit. If the electron is observed only six times out of ten trials, the probability of finding the electron is 0.6. It is not true if we say only 60% of the electron is observed all the time. Then one has to make the probability of finding the electron to be equal to unity by multiplying the wave function with an appropriate constant.

“The process of making the probability density to be equal to unity is called normalization and the constant of multiplication is called normalization constant”. The above explanation can be expressed in terms of equations as

$$\text{Let } \int_{-\infty}^{\infty} \Psi^* \Psi dx = N$$

$$\text{Then } \frac{1}{N} \int_{-\infty}^{\infty} \Psi^* \Psi dx = 1$$

$$\text{Or } \int_{-\infty}^{\infty} \frac{1}{\sqrt{N}} \Psi^* \cdot \frac{1}{\sqrt{N}} \Psi dx = 1 \quad \text{----- (i)}$$

Where $\left(\frac{1}{\sqrt{N}}\right)$ the normalization constant and the wave function is $\frac{1}{\sqrt{N}} \Psi$ is called normalized wave function. If we denote $\frac{1}{\sqrt{N}} \Psi$ by new wave function ψ equation (i) becomes

$$\int_{-\infty}^{\infty} \psi^* \psi dx = 1 \quad \text{----- (ii)}$$

This is the condition for normalization. Any wave function satisfying above equation is said to be normalized.

Setting up of 1-D time independent Schrodinger's Wave Equation

Consider a particle of mass m , moving with a velocity v and a momentum p along +ve X -axis. The de Broglie waves associated with the moving particle have a wavelength in the neighbourhood of

$$\lambda = \frac{h}{p} \quad \text{----- (1)}$$

The wave function representing those de Broglie waves is of the form

$$\Psi = A e^{i(\omega t - kx)} \quad \text{----- (2)}$$

Where ω is the angular frequency and K is the Wave number.

Differentiating the above equation twice partially w.r.t. x , we get

$$\begin{aligned} \frac{d\Psi}{dx} &= -ik.A e^{i(\omega t - kx)} \\ \frac{d^2\Psi}{dx^2} &= -K^2 A e^{i(\omega t - kx)} \\ \frac{d^2\Psi}{dx^2} &= -K^2 \Psi \\ \therefore \frac{d^2\Psi}{dx^2} + K^2 \Psi &= 0 \quad \text{----- (3)} \end{aligned}$$

$$\begin{aligned} \text{We know that the wave number } k &= \frac{2\pi}{\lambda} \Rightarrow K^2 = \frac{4\pi^2}{\lambda^2} \\ \therefore \frac{d^2\Psi}{dx^2} + \frac{4\pi^2}{\lambda^2} \Psi &= 0 \end{aligned}$$

$$\begin{aligned} \text{From equation (1), we have } \lambda^2 &= \frac{h^2}{p^2} \Rightarrow \frac{1}{\lambda^2} = \frac{p^2}{h^2} \\ \therefore \frac{d^2\Psi}{dx^2} + \frac{4\pi^2 p^2}{h^2} \Psi &= 0 \quad \text{----- (4)} \end{aligned}$$

$$\text{We have, kinetic energy } E_k = \frac{1}{2} m v^2 = \frac{p^2}{2m} \Rightarrow p^2 = 2m E_k$$

$$\therefore \frac{d^2\Psi}{dx^2} + \frac{8\pi^2 m E_k}{h^2} \Psi = 0 \quad \text{----- (5)}$$

The total energy $E = E_K + V \Rightarrow E - V = E_K$

$$\therefore \frac{d^2 \Psi}{dx^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0 \quad \text{----- (6)}$$

This is the time independent Schrodinger's wave equation in 1-D.

Note:

(i) For a free particle $V = 0$.

$\therefore$ 1-D time independent Schrodinger's wave equation for a free particle is

$$\frac{d^2 \Psi}{dx^2} + \frac{8\pi^2 m E}{h^2} \Psi = 0$$

(ii) The time independent Schrodinger's wave equation for three dimensions is

$$\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

Eigen Value equation - Eigen values and Eigen functions:

On rearranging S. W. E.

$$\left[-\frac{h^2}{8\pi^2 m} + V \right] \Psi = E \Psi$$

I.e. $\hat{H} \Psi = E \Psi$ Where $\left[-\frac{h^2}{8\pi^2 m} + V \right] = \hat{H}$ where $\hat{H}$ is called Hamiltonian Operator.

Above equation is called Eigen value equation. Any wave function satisfying this equation is called eigen function.

An operator operating on a wave function gives constant times the same wave function. The constant thus obtained is called Eigen value of the operator. Eigen value of an operator refers to the expectation value (most probable value) of the dynamical variable represented by that particular operator.

The commonly used operators and the corresponding dynamical variables are listed below.

Operators in Quantum Mechanics:

Operator is a mathematical rule which yield the expectation value of a dynamical variable.

Eg. Consider the function $y = \sin 2x$, if $\left(\frac{d}{dx} \right)$ is the operator then

$$\frac{d}{dx} (\sin 2x) = 2 \cos 2x \text{ which implies } \sin 2x \text{ is not the Eigen function of the operator } \left(\frac{d}{dx} \right).$$

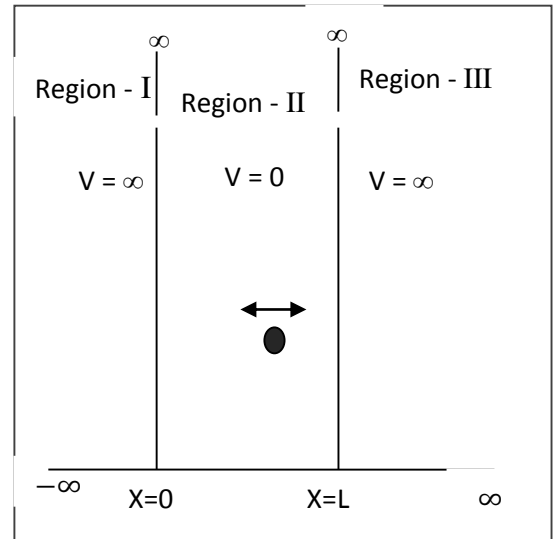
Now consider the operator $\left(\frac{d^2}{dx^2} \right)$, then

$$\frac{d^2}{dx^2} (\sin 2x) = -4 \sin 2x \text{ which implies } \sin 2x \text{ is the Eigen function of the operator } \left(\frac{d^2}{dx^2} \right) \text{ and } -4 \text{ is the eigen value of } \left(\frac{d^2}{dx^2} \right).$$

PARTICLE IN A POTENTIAL WELL OF INFINITE HEIGHT AND FINITE WIDTH:

The Wave Function and Energy Eigen Values:

Consider a moving free particle trapped between imaginary walls of infinite height at $x=0$ and $x = a$. The potential acting on the particle inside the well is zero. The potential outside the well is ∞ . The dynamical behaviour of the particle is studied by solving the time independent Schrodinger's wave equation.



For the region I and III, S.W.E can be written as

$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - \infty) \Psi = 0 \quad \text{----- (1)}$$

The only possible solution for the above equation is $\Psi = 0$, i.e. the particle cannot enter in to the region I and III.

For the region II, S.W.E can be written as

$$\frac{d^2\Psi}{dx^2} + \frac{8\pi^2mE}{h^2} \Psi = 0 \quad \text{----- (2)}$$

Put $\frac{8\pi^2mE}{h^2} = K^2 \quad \text{----- (2a)}$

Then $\frac{d^2\Psi}{dx^2} + K^2 \Psi = 0$

The solution to the above equation is of the form $\Psi = C \cos Kx + D \sin Kx \quad \text{----- (3)}$

The behaviour of the particle at the walls forms the boundary conditions. As it is continuously in motion in region II, the particle cannot stick on to the walls.

- $\therefore$ The boundary conditions are (i) $\Psi=0$ at $x=0$ and
(ii) $\Psi=0$ at $x = a$.

From the 1st boundary condition $\Psi_{x=0} = C \cos Kx_{x=0} + D \sin Kx_{x=0}$
 $0 = C + 0$
 $\Rightarrow C = 0$

The wave function becomes $\Psi = D \sin Kx \quad \text{----- (4)}$

From the 2nd boundary condition $\Psi_{x=a} = D \sin Ka$
 $0 = D \sin Ka$

Since $D \neq 0$, $\sin Ka = 0$

$\Rightarrow Ka = \sin^{-1}(0)$

$\therefore Ka = n\pi$, where $n = 0, 1, 2, 3, \dots$

And $K = \frac{n\pi}{a} \quad \text{----- (4a)}$

Therefore the wave function becomes $\Psi_n = D \sin\left(\frac{n\pi x}{a}\right)$ where $n = 1, 2, 3, \dots \quad \text{----- (5)}$

As right hand side involves an integral integer, the suffix n is assigned to the wave function also. $n=0$ is excluded because $\Psi_n = 0$ means the particle is nowhere present in region II (which is a contradict).

From (2a), we have $\frac{8\pi^2 mE}{h^2} = K^2$

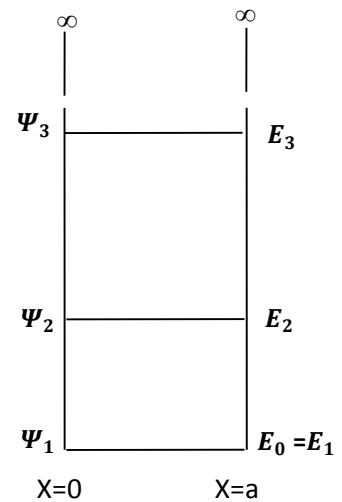
Fro (4a), we have $K = \frac{n\pi}{a}$

$\therefore$ We can write $\frac{8\pi^2 mE}{h^2} = \left(\frac{n\pi}{a}\right)^2$

On re arranging, we get $E = \frac{n^2 h^2}{8ma^2}$

Or $E_n = \frac{n^2 h^2}{8ma^2}$ where $n = 1, 2, 3, \dots$ ----- (6)

Again, as right hand side involves an integral integer, the suffix n is assigned to the energy also. $n=0$ is excluded because $E_n = 0$ means the particle not at all moving in region II (which is contradict).



The least possible energy of the particle inside the potential well is called **Zero point energy** or **ground state energy**. All other possible energies (E_2 , E_3 , and so on, called excited states) are the integral multiple of this ground state energy. Therefore the energy of the particle confined in a potential well is said to be quantized. All the characteristic energy levels are represented by appropriate wave functions, called eigen functions. The figure shows the energy eigen values and corresponding eigen functions of the particle confined in an infinite potential well and finite width.

Normalization of the wave function:

We know that the condition for normalization is $\int_{-\infty}^{\infty} \Psi^* \Psi dx = 1$ ----- (7)

But , we know that the wave function Ψ_n is zero in region I and region III and is real only in region II.

Therefore, the condition for normalization reduces to $\int_0^a \Psi_n^* \Psi_n dx = 1$ ----- (8)

Where $\Psi_n = D \sin\left(\frac{n\pi x}{a}\right)$; $\Psi_n^* = D^* \sin\left(\frac{n\pi x}{a}\right)$

$$\therefore \int_0^a \Psi_n^* \Psi_n dx = \int_0^a D^* \sin\left(\frac{n\pi x}{a}\right) \cdot D \sin\left(\frac{n\pi x}{a}\right) dx = 1$$

$$D^* \cdot D \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) dx = 1$$

$$|D|^2 \cdot \frac{1}{2} \int_0^a \left[1 - \cos\left(\frac{2n\pi x}{a}\right)\right] dx = 1$$

$$|D|^2 \cdot \frac{1}{2} \{[a - 0] - [0 - 0]\} = 1$$

$$\therefore |D|^2 \cdot \frac{a}{2} = 1$$

$$\text{Or } |D|^2 = \frac{2}{a}$$

$$\text{Or } D = \sqrt{\frac{2}{a}}$$

Therefore the normalized wave function for the particle confined in an infinite potential well and finite width is $\Psi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$ ----- (9)

+Eigen functions, probability densities and energy levels for particle in an infinite potential well:

The wave function and the particle confined in an infinite potential well and finite width is

$$\Psi_n = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad \text{----- (a)}$$

and $E_n = \frac{n^2 h^2}{8ma^2} = n^2 E_0$ where $n = 1, 2, 3, \dots$ ----- (b)

Case I $\rightarrow n=1$: The energy corresponding to $n=1$ is the lowest possible energy, called zero point energy (Ground state energy), denoted by E_0 . This energy level is represented by the wave function

$$\Psi_1 = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)$$

$\Psi_1 = 0$ for $x = 0$ and $x = a$ and Ψ_1 is maximum when $x = a/2$.

The wave function Ψ_1 and the probability density $|\Psi_1|^2$ can be represented as shown in figure.

Case II $\rightarrow n=2$: The energy corresponding to $n = 2$ is $E_2 = 4E_0$ called the 1st excited state. This energy level is represented by the wave function $\Psi_2 = \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)$

$\Psi_2 = 0$ for $x = 0, a/2$ and a .

$\Psi_2 = \sqrt{\frac{2}{a}}$ for $x = a/4$ and $\Psi_2 = -\sqrt{\frac{2}{a}}$ for $x = 3a/4$.

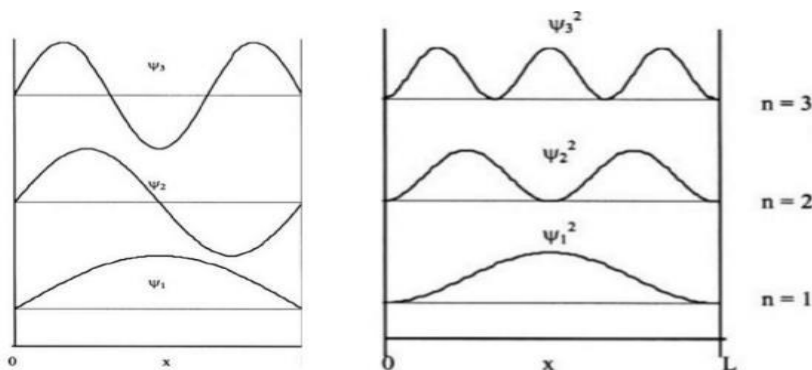
The wave function Ψ_2 and the probability density $|\Psi_2|^2$ can be represented as shown in figure.

Case III $\rightarrow n=3$: The energy corresponding to $n = 3$ is $E_3 = 9E_0$ called the 2nd excited state. This energy level is represented by the wave function $\Psi_3 = \sqrt{\frac{2}{a}} \sin\left(\frac{3\pi x}{a}\right)$

$\Psi_3 = 0$ for $x = 0, a/3$ and $2a/3$.

$\Psi_3 = \sqrt{\frac{2}{a}}$ for $x = a/6$ & $5a/6$ and $\Psi_3 = -\sqrt{\frac{2}{a}}$ for $x = a/2$.

The wave function Ψ_3 and the probability density $|\Psi_3|^2$ can be represented as shown in figure given below.



Need for Quantum Statistics

Classical statistics due to Maxwell-Boltzmann explained the energy and velocity distribution of the molecules of an ideal gas but it failed to explain the observed energy distribution of electrons in the so called '*electron gas*' and that of photons in the '*photon gas*'.

Moreover, M-B distribution function suffers from the following two major objections –

- i. First the particles are assumed to be distinguishable though electrons or other elementary particles are indistinguishable.
- ii. Any number of particles was allowed to occupy the same quantum state while many particles particularly electrons, obey Pauli's exclusion principle which does not allow a quantum state to accept more than one particle.

In classical statistics, $\frac{n_i}{g_i} \ll 1$

So, the probability of a cell containing two or more particles is negligible and the particles thus can be treated as distinguishable.

But, all natural systems obey uncertainty principle and minimum size of a cell in phase space in such a case is h^3 so that number of cells g_i is limited by the value of h^3 .

When occupation index $\frac{n_i}{g_i} \approx 1$, the particles cannot be treated as distinguishable and thus classical statistics cannot therefore be applied.

These difficulties were resolved by the use of Quantum statistics and can be divided as:

- i. **Bose-Einstein (BE) statistics**
- ii. **Fermi-Dirac (FD) statistics**

Fermi-Dirac (FD) Statistics

The basic assumptions of FD statistics are:

- It is applicable for *identical* and *indistinguishable* particles.
- Total energy and total number of the particles of the entire system is constant.
- Spin of the particles should be half integral spin i.e. $\frac{1}{2}\hbar, \frac{3}{2}\hbar, \dots$
- Particles must obey Pauli's exclusion principle, the number of particles in each sublevel will be one or more i.e. $g_i \geq n_i$.
- Functions associated with the particles should be anti-symmetric.
- All particles which obey F.D. statistics are known as *Fermions* (*electron, proton, neutron, ^3He , neutrino, muons, all hyperons ($\Lambda, \Sigma, \Xi, \Omega$) etc.*).

Fermi-Dirac distribution law

Consider a system of Fermions having total number of particles N . We assume that these particles can be divided into different quantum groups or levels having energies $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_k$ and the number of particles in these levels are $n_1, n_2, \dots, n_k$, respectively and let g_i represent the degeneracy of the energy level ε_i .

Since the particles are indistinguishable and only one fermion can accommodate in one cell.

So, the number of meaningful ways in which n_i particles are arranged in g_i quantum states is given by - ${}^{g_i}C_{n_i}$

$$\frac{(g_i)(g_i - 1) \dots (g_i - n_i + 1)}{n_i!} \quad \text{or} \quad \frac{g_i!}{n_i!(g_i - n_i)!}$$

Thus, the total thermodynamic probability for an assembly of fermions (or particular macrostate) is

$$W(n_1, n_2 \cdots n_k) = \prod_{i=1}^k \frac{g_i!}{n_i!(g_i - n_i)!}$$

Most probable state:

taking natural logarithms

$$\ln(W) = \sum_{i=1}^k \ln(g_i!) - \sum_{i=1}^k \ln(n_i!) - \sum_{i=1}^k \ln((g_i - n_i)!)$$

Using Stirling's approximation in above equation, we get

$$\ln(W) = \sum_{i=1}^k \ln(g_i!) - \sum_{i=1}^k n_i \ln(n_i) + \sum_{i=1}^k n_i - \sum_{i=1}^k (g_i - n_i) \ln(g_i - n_i) + \sum_{i=1}^k (g_i - n_i)$$

Differentiating on both sides, we get

$$d(\ln(W)) = -\sum_{i=1}^k dn_i \ln(n_i) + \sum_{i=1}^k (dn_i) \ln(g_i - n_i)$$

For the most probable state, $d(\ln(W))=0$

$$d(\ln(W)) = \sum_{i=1}^k \ln\left(\frac{g_i - n_i}{n_i}\right)(dn_i)$$

In addition, the system must satisfy two auxiliary conditions:

- i. Conservation of total number of particles: $dn = \sum_{i=1}^k dn_i = 0$
- ii. Conservation of total energy of the system: $dU = \sum_{i=1}^k \varepsilon_i dn_i = 0$

Applying Lagrange's method of undetermined multipliers

$$\sum_{i=1}^k \ln\left(\frac{g_i - n_i}{n_i}\right)(dn_i) - \alpha \sum_{i=1}^k (dn_i) - \beta \sum_{i=1}^k \varepsilon_i (dn_i) = 0$$

$$\sum_{i=1}^k \left[\ln \left(\frac{g_i - n_i}{n_i} \right) - \alpha - \beta \varepsilon_i \right] (dn_i) = 0$$

Proceeding as before

$$\ln \left(\frac{g_i - n_i}{n_i} \right) - \alpha - \beta \varepsilon_i = 0 \Rightarrow \ln \left(\frac{g_i - n_i}{n_i} \right) = \alpha + \beta \varepsilon_i$$

$$\frac{g_i}{n_i} - 1 = e^{\alpha + \beta \varepsilon_i} \Rightarrow \boxed{f(\varepsilon_i) = \frac{n_i}{g_i} = \frac{1}{e^{\alpha + \beta \varepsilon_i} + 1}} \quad \text{FD distribution function}$$

Now, as we know that $\beta = \frac{1}{kT}$

The other multiplier, α is related to the chemical potential μ

$$\alpha = \frac{-\mu}{kT} = \frac{-E_F}{kT} \quad \text{where } E_F \text{ is Fermi energy}$$

Bose-Einstein (BE) Statistics

The basic assumptions of BE statistics are:

- It is applicable for *identical* and *indistinguishable* particles.
- Any number of particles can occupy a single cell in the phase space.
- The size of cell can not be less than h^3 where h is Planck's constant.
- The number of phase space cells is comparable with the number of particles i.e., occupation index, $n_i/g_i=1$.

- It is applicable to particles with integral spin angular momentum in units of $h/2\pi$. All particles which obey B.E. statistics are known as *Bosons* (photon, phonon, all mesons etc).
- The wave function of the system is symmetric under the positional exchange of any two particles.
- Total energy and total number of particles of the entire system is constant.

Bose-Einstein distribution law

Let us consider a system of bosons having total number of particles N . Suppose these identical, indistinguishable and non-interacting particles are to be distributed among g_i different quantum states each having energy ε_i .

So, the number of meaningful arrangements for which n_i indistinguishable particles are to be distributed among g_i cells without any restriction on the number of particles is given by -

$$\text{Number of ways} = \frac{(n_i + g_i - 1)!}{n_i!(g_i - 1)!}$$

So, the thermodynamic probability for a macrostate is then:

$$W(n_1, n_2, \dots, n_k) = \prod_{i=1}^k \frac{(n_i + g_i - 1)!}{n_i!(g_i - 1)!}$$

Taking natural logarithm:

$$\ln(W) = \sum_{i=1}^k \ln((n_i + g_i - 1)!) - \sum_{i=1}^k \ln(n_i!) - \sum_{i=1}^k \ln((g_i - 1)!)$$

Using Stirling's approximations, we get

$$\begin{aligned} \ln(W) = & \sum_{i=1}^k (n_i + g_i - 1) \ln(n_i + g_i - 1) - \sum_{i=1}^k (n_i + g_i - 1) - \sum_{i=1}^k n_i \ln(n_i) \\ & + \sum_{i=1}^k n_i - \sum_{i=1}^k (g_i - 1) \ln(g_i - 1) + \sum_{i=1}^k (g_i - 1) \end{aligned}$$

On further solving, we get

$$d(\ln(W)) = - \sum_{i=1}^k \ln\left(\frac{n_i}{n_i + g_i}\right) dn_i$$

Using Lagrange's method of undetermined multipliers wherein we multiply with two constraints α and β , respectively, one get

$$\sum_{i=1}^k \ln\left(\frac{n_i}{n_i + g_i}\right) dn_i + \alpha \sum_{i=1}^k dn_i + \beta \sum_{i=1}^k \varepsilon_i dn_i = 0$$

Proceeding as before:

$$\ln\left(\frac{n_i}{n_i + g_i}\right) + \alpha + \beta\varepsilon_i = 0 \quad \text{since } dn_i \neq 0$$

Since, n_i and g_i are very large therefore, $n_i + g_i - 1 = n_i + g_i$

$$\ln\left(\frac{n_i}{n_i + g_i}\right) = \exp[-(\alpha + \beta\varepsilon_i)]$$

$$\frac{g_i + n_i}{n_i} = \exp(\alpha + \beta\varepsilon_i)$$

$$\frac{g_i}{n_i} = e^{\alpha + \beta\varepsilon_i} - 1 \quad \Rightarrow \quad \frac{n_i}{g_i} = \frac{1}{e^{\alpha + \beta\varepsilon_i} - 1}$$

Hence, expression for the most probable distribution of particles among various energy levels of a system obeying B-E statistics is given by -

$$\boxed{n_i = \frac{g_i}{e^{\alpha + \beta\varepsilon_i} - 1}} \quad (\text{Bose - Einstein distribution law})$$

In general, combining all the three statistics, we can write the expression as

$$f(\varepsilon) = \frac{n_i}{g_i} = \frac{1}{e^{(\varepsilon_i - \mu)/kT} + K}$$

where, constant K takes values 0, -1 and 1 for M-B (classical), B-E and F-D statistics.

Planck's Derivation of the Energy Density of Blackbody Radiation

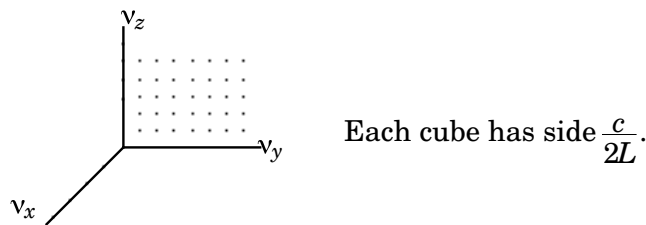
To calculate the number of modes of oscillation of electromagnetic radiation possible in a cavity, consider a one-dimensional box of side L . In equilibrium only standing waves are possible, and these will have nodes at the ends $x = 0, L$.

$$\frac{L}{\lambda} = \frac{n_x}{2} \quad n_x = 1, 2, \dots$$

and since $\lambda\nu = c = \text{speed of propagation for all wave motion}$,

$$\nu = n_x \frac{c}{2L}$$

There will be two modes for each trio of integers (one for each dimension) n_x, n_y, n_z because there are two independent polarizations possible. To find the number of modes with frequency between ν and $\nu + d\nu$, look at an array of points:



There is one point per cube of volume $\left(\frac{c}{2L}\right)^3$, and only positive integers, n_x, n_y, n_z are acceptable. Thus the number of triplets of positive integers is equivalent to the volume of one octant of the space divided by the volume $\left(\frac{c}{2L}\right)^3$:

$$\text{no. modes of oscillation between } \nu \text{ and } \nu + d\nu = \frac{2 \times \frac{1}{8} \times 4\pi\nu^2 d\nu}{\left(\frac{c}{2L}\right)^3} = \frac{8\pi V}{c^3} \nu^2 d\nu$$

The factor $4\pi\nu^2 d\nu$ is the volume of a thin spherical shell; L^3 has been replaced by V , the volume of the cavity. It is convenient to express this *density of states* in terms of other variables:

$$\begin{aligned} g(\nu) d\nu &= \frac{8\pi V}{c^3} \nu^2 d\nu & g(\lambda) d\lambda &= \frac{8\pi V}{\lambda^4} d\lambda \\ g(\epsilon) d\epsilon &= \frac{8\pi V}{(hc)^3} \epsilon^2 d\epsilon & g(p) dp &= \frac{8\pi V}{h^3} p^2 dp \end{aligned}$$

$$\epsilon = h\nu = pc$$

The expressions involving frequency ν , energy ϵ , and wavelength λ are classical physics. If we assume that each mode of oscillation represents a harmonic oscillator, with $\frac{1}{2}kT$ each potential and kinetic energy on the average (in accordance with the equipartition theorem), we get the *Rayleigh-Jeans law*:

$$\frac{\text{Energy}}{\text{Volume}} \equiv u_\nu d\nu = \frac{8\pi}{c^3} kT \nu^2 d\nu \quad \text{or} \quad \frac{\text{Energy}}{\text{Volume}} \equiv u_\lambda d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$$

The divergence of this relation at high frequency or low wavelength was known as the *ultraviolet catastrophe*. Planck's new idea was to assume that the possible energies of the oscillators were quantized, *i.e.*, that oscillators of frequency ν could only have energy

$$\epsilon_n = nh\nu \quad n = 0, 1, 2, \dots$$

where h was a new constant he introduced. Now known as *Planck's constant*, it was determined by fitting the theoretical curve to the experimental data. The average energy per oscillator was calculated from the Maxwell-Boltzmann distribution:

$$\bar{\epsilon} = \frac{\sum_n \epsilon_n e^{-\epsilon_n/kT}}{\sum_n e^{-\epsilon_n/kT}}$$

The denominator is called the *partition function*, and is often represented by Z . It is easily evaluated by summing the geometric series:

$$Z = \sum_{n=0}^{\infty} e^{-\epsilon_n/kT} = \sum_{n=0}^{\infty} e^{-nx} = \frac{1}{1 - e^{-x}} \quad \text{where } x = \frac{h\nu}{kT}$$

The numerator can then be found from the denominator:

$$\sum_{n=0}^{\infty} n h\nu e^{-nx} = h\nu \left(-\frac{dZ}{dx} \right) = \frac{h\nu e^{-x}}{(1 - e^{-x})^2}$$

and the average energy per oscillator is seen to be

$$\bar{\epsilon} = \frac{h\nu}{e^x - 1} = \frac{h\nu}{e^{h\nu/kT} - 1}$$

Thus the energy per unit volume of the radiation in the cavity is

$$u_{\nu}(T) d\nu = \frac{8\pi}{c^3} \frac{h\nu^3}{e^{h\nu/kT} - 1} d\nu \quad \text{or} \quad u_{\lambda}(T) d\lambda = \frac{8\pi hc}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1} d\lambda$$

The total energy per unit volume (energy density) is the integral over all frequencies or wavelengths:

$$u(T) = \frac{8\pi h}{c^3} \int_0^{\infty} \frac{\nu^3}{e^{h\nu/kT} - 1} d\nu = \frac{8\pi (kT)^4}{(hc)^3} \int_0^{\infty} \frac{x^3}{e^x - 1} dx$$

The integral is obviously a pure number. It happens to be $\pi^4/15$. Thus the energy density in a black body is

$$u(T) = \frac{8\pi^5 (kT)^4}{15(hc)^3} \equiv aT^4$$

This may be thought of as one form of the Stefan-Boltzmann law. [Josef Stefan in 1879 showed experimentally that the flux from a cavity in thermal equilibrium is proportional to the fourth power of the absolute temperature, and Ludwig Boltzmann in 1884 derived this fourth power relation from thermodynamic theory. Until Planck's work, there was no theoretical method of determining the constants of proportionality.]

The *flux* radiated from the surface of a black body is related to the energy density:

$$F_{\nu} = \frac{c}{4} u_{\nu} = \frac{2\pi}{c^2} \frac{h\nu^3}{e^{h\nu/kT} - 1} \quad \text{or} \quad F_{\lambda} = \frac{c}{4} u_{\lambda} = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

where $F_{\nu} d\nu = \text{flux} = \frac{\text{energy}}{\text{area} \cdot \text{time}}$ with frequency between ν and $\nu + d\nu$.

The total flux, obtained by integrating over all frequencies, is

$$F = \frac{c}{4} u = \frac{ac}{4} T^4 \equiv \sigma T^4$$

This is the usual form of the Stefan-Boltzmann law. The constant

$$\sigma = 5.670 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4} = 5.670 \times 10^{-5} \frac{\text{erg}}{\text{cm}^2 \text{s K}^4} = \text{Stefan-Boltzmann constant.}$$

It is of interest to look at the limits of the Planck distribution.

At low frequency or large wavelength,

$$u_{\nu}(T) \rightarrow \frac{8\pi\nu^2 kT}{c^3} \quad \text{and} \quad u_{\lambda}(T) \rightarrow \frac{8\pi kT}{\lambda^4} = \text{Rayleigh-Jeans law.}$$

Note that Planck's constant drops out. This is one example of the correspondence principle; as h becomes negligible compared to other quantities in the quantum mechanical law, the result approaches the classical law. At high frequency or small wavelength,

$$u_{\nu}(T) \rightarrow \frac{8\pi h\nu^3}{c^3} e^{-h\nu/kT} \quad \text{and} \quad u_{\lambda}(T) \rightarrow \frac{8\pi hc}{\lambda^5} e^{-hc/\lambda kT}$$

The frequency or wavelength of maximum flux can be found by setting the derivative with respect

to ν or λ equal to zero:

$$0 = \frac{d}{d\nu} F_\nu = \frac{2\pi h}{c^2} \frac{1}{e^{h\nu/kT} - 1} \left[3\nu^2 - \frac{\nu^3 (h/kT) e^{h\nu/kT}}{e^{h\nu/kT} - 1} \right]$$

which simplifies to $\left[3 = \frac{xe^x}{e^x - 1} \right]$ with $x = \frac{h\nu}{kT}$

The transcendental equation may be solved graphically [graph $y = x$ and $y = 3(1 - e^{-x})$ and find the nonzero point of intersection] or numerically. The result is $x = 2.821$ meaning that flux as a function of frequency is a maximum at $\nu_{\max} = 2.821 kT/h$. Maximization of F_λ is similar:

$$0 = \frac{d}{d\lambda} F_\lambda \Rightarrow \frac{ye^y}{e^y - 1} = 5 \quad \text{with } y = \frac{hc}{\lambda kT} \quad \text{which yields } y = 4.965.$$

Flux as a function of wavelength is a maximum at

$$\lambda_{\max} = \frac{hc}{4.965kT} = \frac{2.90 \text{ mm} \cdot \text{K}}{T}.$$

At room temperature $T \approx 290 \text{ K}$, and thermal radiation is a maximum at $\lambda \approx 0.01 \text{ mm} = 10 \mu\text{m}$, in the infrared. Fortunately, our eyes are not sensitive to this wavelength. The maximum intensity of the sun's radiation is at $\lambda \approx 500 \text{ nm}$, implying that the sun's *surface temperature* is $T \approx 5800 \text{ K}$. The variation of intensity with wavelength of the sun [and other stars] is not exactly that of a black body, but it is rather close. The universal microwave background radiation, peaked at $\lambda \approx 1 \text{ mm}$, fits the Planck curve for a black body of $T = 2.728 \text{ K}$ to great precision. (The deviation, of order 6 parts in 10^6 is, of course, of great interest.)

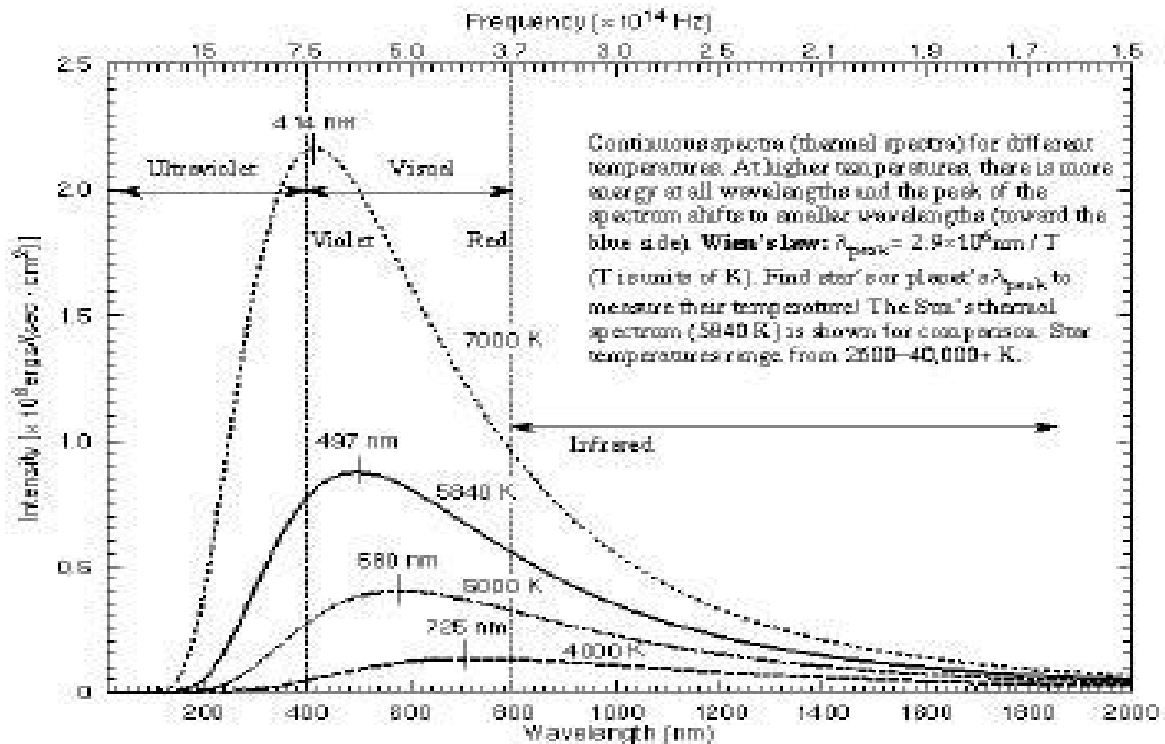


Illustration source: <http://www.bc.cc.ca.us/programs/sea/astronomy/light/lightb.htm#A2.1>

Now at <http://www.astronomynotes.com/light/s4.htm>

Scanning tunneling microscope (STM)

A scanning tunneling microscope (STM) is a non-optical microscope that works by scanning an electrical probe tip over the surface of a sample at a constant spacing. This allows a 3D picture of the surface to be created.

Scanning tunneling microscope (STM) is widely used in both industrial and fundamental research to obtain atomic-scale images of metal surfaces.

It provides a three-dimensional profile of the surface which is very useful for characterizing surface roughness, observing surface defects, and determining the size and conformation of molecules and aggregates on the surface.

Scanning tunneling microscope

An ordinary microscope, which employs optical lenses, could view objects smaller than the wavelength of light. An electron microscope could view smaller things with greater clarity than an optical microscope, but still could not clearly view individual atoms.

So Binnig and Rohrer decided to build their own instrument – something new that would be capable of seeing and manipulating atoms at the nanoscale level.

To do that, they began experimenting with tunneling, a quantum phenomenon in which atoms escape the surface of a solid to form a kind of cloud that hovers above the surface; when another surface approaches, its atomic cloud overlaps and an atomic exchange occurs.

Principle of Scanning tunneling microscope

STM is a type of microscope whose principle of operation is based on the [quantum](#) mechanical phenomenon known as [tunneling](#), in which the wavelike properties of [electrons](#) permit them to “tunnel” beyond the surface of a solid into regions of space that are forbidden to them under the rules of classical [physics](#).

The probability of finding such tunneling electrons decreases exponentially as the distance from the surface increases.

Electrons use a quantum mechanical effect to ‘tunnel’ from the tip to the sample and vice versa.

Working of STM

The STM sample must conduct electricity for the process to work. The STM uses a tip that ends in a single atom, and a voltage is passed through the tip and the sample.

The sharp tip of a [tungsten](#) needle is positioned a few angstroms from the sample surface. A small voltage is applied between the probe tip and the surface, causing electrons to tunnel across the gap.

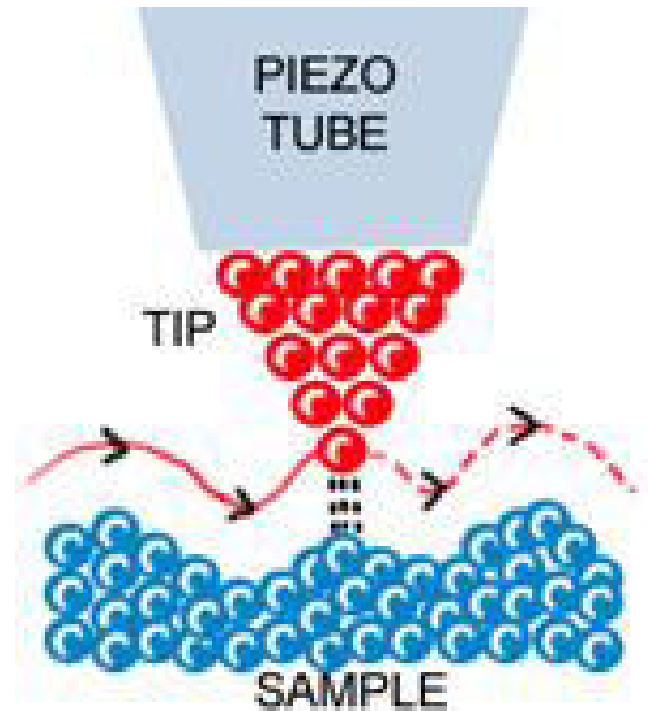
As the probe is scanned over the surface, it registers variations in the tunneling current, and this information can be processed to provide a topographical image of the surface.

Working of STM

The current that results depends upon the distance between the probe tip and the sample surface.

The tip is attached to a piezoelectric tube, and the voltage applied to the piezo rod is altered to maintain a constant distance of the tip from the surface.

Changes in this voltage allow a three-dimensional picture of the material surface to be built up as the tip is scanned back and forth across the sample.

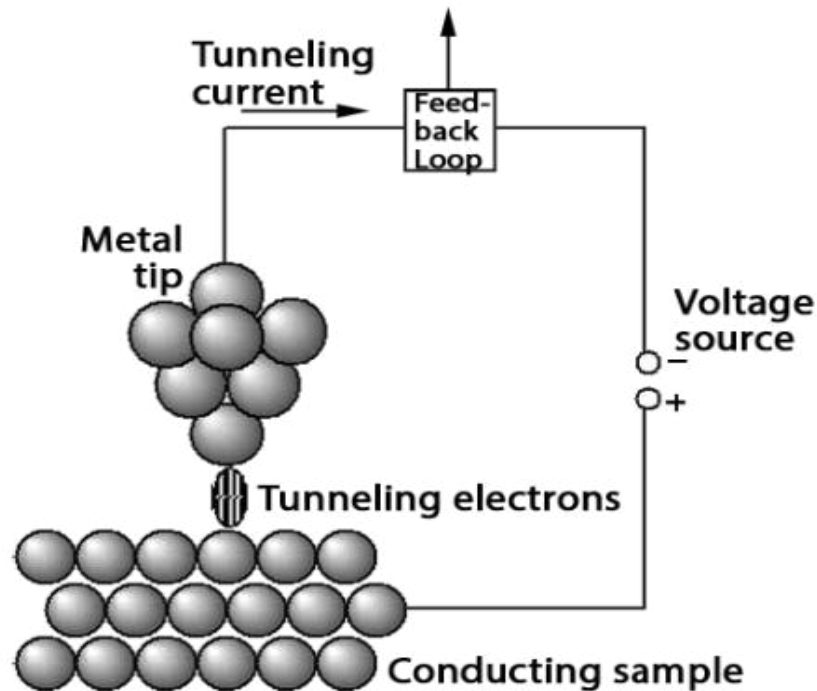


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Working of STM



Feedback loop and electron tunneling for scanning tunneling microscopy

Piezoelectric Effect

The piezoelectric effect was discovered by Pierre Curie in 1880. The effect is created by squeezing the sides of certain crystals, such as quartz or barium titanate.

The result is the creation of opposite charges on the sides. The effect can be reversed as well; by applying a voltage across a piezoelectric crystal, it will elongate or compress.

These materials are used to scan the tip in an scanning tunneling microscopy (STM) and most other scanning probe techniques. A typical piezoelectric material used in scanning probe microscopy is PZT (lead zirconium titanate).

Feedback Loop

Electronics are needed to measure the current, scan the tip, and translate this information into a form that we can use for STM imaging.

A feedback loop constantly monitors the tunneling current and makes adjustments to the tip to maintain a constant tunneling current.

These adjustments are recorded by the computer and presented as an image in the STM software. Such a setup is called a constant current image.

In addition, for very flat surfaces, the feedback loop can be turned off and only the current is displayed. This is a constant height image.

Constant Current and Height Mode

By using a feedback loop the tip is vertically adjusted in such a way that the current always stays constant. As the current is proportional to the local density of states, the tip follows a contour of a constant density of states during scanning. A kind of a topographic image of the surface is generated by recording the vertical position of the tip.

In this mode the vertical position of the tip is not changed, equivalent to a slow or disabled feedback. The current as a function of lateral position represents the surface image. This mode is only appropriate for atomically flat surfaces as otherwise a tip crash would be inevitable.

Resolution

An STM can have a lateral resolution of up to 0.1nm, and a depth resolution of up to 0.01nm, which combined is small enough to resolve individual atoms.

In recent years, carbon nanotubes have been used for constructing these tips, allowing for improved resolution and reduced error in image reconstruction.

Summary

STM is a type of microscope whose principle of operation is based on the [quantum](#) mechanical phenomenon known as [tunneling](#), in which the wavelike properties of [electrons](#) permit them to “tunnel” beyond the surface of a solid into regions of space that are forbidden to them under the rules of classical [physics](#)